

Station challenge cards (predict first)

Read the card, write your prediction FIRST, then test it with the slinky and record what happened. Keep the slinky flat on the floor, never let go while it is stretched, and wear goggles while it is stretched.

1 Count the round trips

Send one sharp pulse down the stretched slinky. Predict how many times it will travel down and back before it fades out, then send it and count.

My prediction: _____ What I measured: _____

2 Tension change

One teammate pulls the slinky a little tighter (do not overstretch). Predict whether the pulse comes back faster or slower than before, then time it.

My prediction: _____ What I measured: _____

3 Pinch a wall

A teammate pinches the middle of the slinky to make a fixed point. Predict what the pulse does when it reaches the pinch, then send one and watch.

My prediction: _____ What I measured: _____

4 Measure the speed

Time N full down-and-back round trips on a measured lane of length L. Total distance = $2 \times L \times N$. Divide that distance by the total time and record the speed in metres per second.

My prediction: _____ What I measured: _____

5 Two pulses meet

Two teammates each send a pulse from opposite ends at the same time. Predict what happens when the two pulses meet in the middle, then try it.

My prediction: _____ What I measured: _____

6 Big push or small push

Send a small pulse, then a big one along the same lane. Predict whether the big push travels faster, slower, or the same speed, then time both.

My prediction: _____ What I measured: _____